DECENTRALISED MULTI-AGENT SEARCH AND RESCUE UNDER LIMITED COMMUNICATION

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ABSTRACT

A person is trapped at an unknown location inside a hazardous building, and a team of heterogeneous autonomous agents must locate them and bring them to an exit. Search drones sense well but cannot perform the rescue; ground rescue robots can perform the rescue but sense poorly, so information must travel from the agents that locate the victim to the agents that can reach them. Communication, however, is a limited resource. This project studies that trade-off. The building is modelled as a graph with obstacles, hazardous locations, an exit, and one unknown victim location; agents act on local, noisy observations under a communication budget. Task allocation is modelled using constraint satisfaction and distributed constraint optimisation, and a centralised task-allocation baseline is compared with a decentralised DCOP-based method in simulation. Experiments will vary the communication budget, sensor reliability, team size, and building complexity, and performance will be measured by rescue success rate, completion time, communication cost, and movement cost. No experimental results are reported at this stage.

Keywords Multi-agent systems · Search and rescue · Decentralised coordination · Task allocation · Constraint satisfaction · Distributed constraint optimisation · Limited communication

1 Introduction

Search-and-rescue operations in hazardous buildings are difficult because responders often do not know where a trapped person is, or which routes remain safe. Autonomous robots can support such operations by exploring, sensing, and mapping, and multi-robot teams have been studied extensively for exactly this setting, covering planning, coordination, and perception for collaborative search and rescue [1]. Teams that mix robots with different capabilities have also been studied for urban search and rescue, where the question of which robot should take which task becomes central [3].

This project considers a team of heterogeneous autonomous agents inside a simulated hazardous building. The team contains *search drones*, which move quickly and sense over a wide area but cannot physically rescue anyone, and *ground rescue robots*, which move only along accessible ground routes and sense poorly but can reach the victim and bring them to an exit. This split is deliberate and is the source of the coordination problem: the agents that find the victim are never the agents that can rescue the victim, so a report has to travel from one group to the other before the task can be completed.

Three properties make this a multi-agent systems problem rather than a path-planning problem. First, no agent observes the whole environment; each acts on local and imperfect observations. Second, capabilities are complementary and non-interchangeable, so tasks must be allocated to agents that are actually able to perform them, which is the classical multi-robot task allocation question [4]. Third, the communication that carries information between agents is a

limited resource, so agents must decide not only where to go but also what is worth saying. Coordination of this kind is commonly modelled as a distributed constraint optimisation problem (DCOP), in which decision variables are held by different agents and solved through local message passing [2].

The main research question is: *how does reducing the communication budget affect the performance of decentralised coordination in multi-agent search and rescue when agents have incomplete and noisy information?* The aim is not to demonstrate that a team of agents can rescue a victim, but to measure the trade-off between communication cost and rescue performance, and to identify the conditions under which decentralised coordination stays close to what a centralised coordinator with pooled information can achieve. A secondary question is whether communication becomes more or less valuable as sensors become less reliable, since sharing observations spreads useful evidence and false alarms alike. A further practical concern is redundant exploration: without communication, several agents may search the same region, an effect well documented in coordinated multi-robot exploration [6].

2 Preliminaries

This section fixes the notation and definitions used in the rest of the report.

Agents and heterogeneity. Let $\mathscr{A}$ denote the set of agents, partitioned into search drones $\mathscr{D}$ and ground rescue robots $\mathscr{R}$, so that $\mathscr{A} = \mathscr{D} \cup \mathscr{R}$ and $\mathscr{D} \cap \mathscr{R} = \emptyset$. Agents are *heterogeneous* when they do not all have the same capabilities. Here, drones have a larger sensing radius and higher sensing reliability, while only ground robots can execute a rescue.

Environment. The building is a graph $G = (V, E)$, where V is a set of locations and E the set of traversable connections between them. In the simulator, V is generated as a 2D grid whose cells are free, obstacles, or hazardous, with one designated exit $v_{\text{exit}} \in V$ and one victim location $v^\star \in V$ that is unknown to the agents. Time is discrete, indexed $t = 1, 2, \ldots$

Incomplete information. An agent's *local observation* is the information it obtains from its own sensors at its current location. Information is *incomplete* when a local observation does not reveal the full state of the environment — here, the victim location and the state of unvisited locations. Observations are additionally *noisy*: a scan may miss a victim that is present, or report a victim where there is none.

Tasks and task allocation. A *task* is an activity an agent may be assigned, such as searching a region or carrying out the rescue. Let $\mathscr{T}$ denote the set of candidate tasks, with $\mathscr{T}_{\text{res}} \subset \mathscr{T}$ the subset of rescue tasks. *Task allocation* is the assignment of tasks to agents subject to their capabilities and to the state of the environment; not every agent can perform every task [4].

Communication. Agents may exchange short messages reporting searched locations or a suspected victim location. The total amount they may exchange during a simulation run is limited by a *communication budget* B. The unit of communication cost, and whether B is charged per agent or to the team, are fixed in the experimental setup.

CSP and DCOP. A *constraint satisfaction problem* (CSP) is a triple of variables, domains, and constraints; a solution assigns to each variable a value from its domain so that all constraints hold. A *distributed constraint optimisation problem* (DCOP) extends this to multiple agents: the variables are distributed among agents, and cost functions defined over small groups of variables express how combinations of decisions affect a shared objective. Agents exchange messages with their neighbours to find an assignment that minimises the total cost [2]. In this project, CSP concepts describe which task assignments are permitted, and the DCOP framework models decentralised task allocation under a communication budget.

3 Problem Formulation

Let $G = (V, E)$, $\mathscr{A} = \mathscr{D} \cup \mathscr{R}$ with $\mathscr{D} \cap \mathscr{R} = \emptyset$, and let $\mathscr{T}$ be the set of candidate tasks with rescue tasks $\mathscr{T}_{\text{res}} \subset \mathscr{T}$. Let $p_i^t \in V$ denote the location of agent $i \in \mathscr{A}$ at time t, and $v^\star \in V$ the victim location, unknown to all agents at $t = 0$.

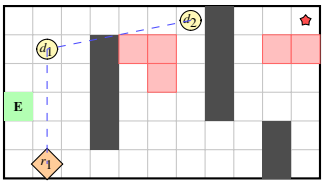


Figure 1: A simulated building instance: obstacles (black), hazardous locations (red), the exit (**E**), search drones d_j, a ground rescue robot r_1, the victim (star, unknown to the agents), and communication links (dashed).

Decision variables. For each agent $i \in \mathscr{A}$ and task $j \in \mathscr{T}$,

$$x_{ij} = \begin{cases} 1, & \text{if task } j \text{ is assigned to agent } i, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Assignment constraints.

$$\sum_{i \in \mathscr{A}} x_{ij} \le 1, \qquad \forall j \in \mathscr{T}, \tag{2}$$

$$x_{ij} \le F_{ij}, \qquad \forall i \in \mathscr{A},\ \forall j \in \mathscr{T}, \tag{3}$$

$$F_{ij} = 0, \qquad \forall i \in \mathscr{D},\ \forall j \in \mathscr{T}_{\text{res}}, \tag{4}$$

where $F_{ij} \in \{0,1\}$ indicates that agent i is capable of performing task j under the prevailing environmental conditions. Constraint (2) allows a task to remain unassigned; (4) encodes the capability split, so rescue tasks are feasible only for ground robots.

Observation model. A scan by agent i returns, for each location v within its sensing radius ρ_i, a binary reading $z_v \in \{0,1\}$ with

$$\Pr(z_v = 1 \mid v = v^\star) = p_i^{\text{det}}, \qquad \Pr(z_v = 1 \mid v \neq v^\star) = p_i^{\text{fa}}, \tag{5}$$

where $p_i^{\text{det}} > p_{i'}^{\text{det}}$ and $\rho_i > \rho_{i'}$ for all $i \in \mathscr{D}$, $i' \in \mathscr{R}$. *Sensor reliability* denotes the pair $(p^{\text{det}}, p^{\text{fa}})$.

Communication constraint. With m_i^t the message sent by agent i at step t, $c(\cdot)$ the per-message cost, and T the final step of a run,

$$C_{\text{comm}} = \sum_{t=1}^{T} \sum_{i \in \mathscr{A}} c\big(m_i^t\big) \le B. \tag{6}$$

Objective. Let T_{rescue} be the completion time, C_{move} the total movement cost over all agents, and $S \in \{0,1\}$ the success indicator, with $S = 1$ if the victim reaches v_{exit} within the step limit and $S = 0$ otherwise; for $S = 0$, T_{rescue} is set to the step limit. The coordination problem is

$$\min_{\{x_{ij}\}} \quad \lambda_t T_{\text{rescue}} + \lambda_c C_{\text{comm}} + \lambda_m C_{\text{move}} + \lambda_f (1 - S), \qquad \lambda_t, \lambda_c, \lambda_m, \lambda_f \ge 0, \tag{7}$$

subject to (2)–(4) and (6). The weights are fixed and reported in the experimental setup.

Decentralised form. The decentralised problem is the DCOP $\langle \mathscr{A}, X, D, F, \alpha \rangle$, where $X = \{x_{ij}\}$ are the decision variables, D their domains $\{0,1\}$, F the set of cost functions derived from (7), and $\alpha : X \to \mathscr{A}$ maps each variable to the agent that controls it, with $\alpha(x_{ij}) = i$. Agent i selects values for $\{x_{ij}\}_{j \in \mathscr{T}}$ using its local observations and messages received from neighbours, so that

$$\min_{\{x_{ij}\}} \sum_{f \in F} f\big(x_f\big) \quad \text{subject to (2)–(4), (6),} \tag{8}$$

where x_f denotes the variables in the scope of cost function f. The centralised problem is the same optimisation with all variables controlled by a single coordinator and C_{comm} not charged.

4 Proposed Solutions

We will implement one simulator of the environment of Section 2 and compare two coordination approaches on identical, randomly generated instances with matched random seeds.

1. **Centralised task-allocation baseline.** A single coordinator receives the agents' observations, maintains one shared view of the environment, and solves the assignment (1)–(4) centrally, then routes each agent to its assigned task. The coordinator is not charged for communication, so this approach is an upper reference on what the available information permits, rather than a competing method.
2. **Decentralised DCOP-based coordination.** Task-allocation decisions are distributed among the agents as in (8). Each agent decides from its own observations and the messages it has received, subject to the budget B. Cost functions penalise two agents being assigned to the same region and reward directing a ground robot toward a reported victim location. Established DCOP algorithms such as ADOPT provide the basis for solving this formulation [5]; the specific algorithm will be selected by the group before implementation, since the object of study is the effect of the communication budget rather than the choice of solver.

The case $B = 0$, in which agents never communicate, will also be run as an ablation separating the benefit of communication from the benefit of having complementary agent types.

4.1 Planned experiments

The **main experiment** varies the communication budget B over four levels — none ($B = 0$), low, moderate, and unrestricted — and reports each metric as a function of the budget, with the centralised baseline as a reference line. A **secondary experiment** repeats this at three levels of sensor reliability (high, medium, low p^{det}), to test whether communication is more or less valuable when observations are noisier. Two further variations, **team size** ($|\mathscr{D}|$ and $|\mathscr{R}|$) and **building complexity** (number of locations, obstacle and hazard density), will be used to check that the observed trend is not specific to one team or one map. In each comparison the remaining conditions are held fixed.

4.2 Metrics

- **Rescue success rate**: proportion of runs with $S = 1$;
- **Completion time**: T_{rescue} on successful runs;
- **Communication cost**: C_{comm}, measured identically for both approaches;
- **Movement cost**: C_{move}, summed over all agents.

4.3 Planned results and expected outcomes

No results are claimed at this stage. By the final submission we plan to report: a curve of rescue performance against communication budget, showing whether performance degrades gradually or drops sharply below some budget; the size of the gap between decentralised coordination and the centralised reference, and the conditions under which that gap widens; whether restricted communication is more damaging when sensors are unreliable; and how much of the team's performance depends on the drone-to-robot handoff, measured against the $B = 0$ ablation. A negative result, such as finding that sharing noisy detections degrades team performance, is a valid outcome provided it is measured and explained.

4.4 Scope and anticipated limitations

Agents may share false detections or disagree about a location, so a simple and clearly stated rule for resolving conflicting reports is required. Under a tight budget, agents cannot tell each other where they have already searched, so some redundant exploration is unavoidable; measuring it is part of the study. Deciding *what* to send is itself a design space, and we restrict it to one simple rule so that the budget, rather than the messaging policy, remains the variable under study. Discrete time, a graph abstraction of the building, a single victim, static hazards, and abstract models of communication and sensor noise are deliberate simplifications that keep the project implementable within the semester. This is a simulation study of coordination under limited communication, not a claim about real emergency-response systems.

5 Contribution

The project will be carried out collaboratively, with each member leading one part and contributing to integration and testing.

- **Boinapally Dhanvini** — literature review on multi-agent search and rescue, heterogeneous-agent coordination, constraint satisfaction, and distributed constraint optimisation; contributes to the formal problem formulation of Section 3, defining agents, candidate tasks, feasibility conditions, and the shared objective, and relates the design to the concepts used in the course.
- **Bollineni Snehitha Rao** — development of the simulated hazardous-building environment: locations, traversable routes, obstacles, hazards, and initial conditions. Implements the heterogeneous agent models, the local observation model (5), sensor-reliability settings, and the building-complexity settings, so that both approaches are evaluated in the same environment.
- **Jils Shah** — implementation of the centralised task-allocation baseline and of the decentralised method based on the DCOP formulation (8). Represents task assignments, capability constraints, and the communication budget in code, integrates both approaches into the simulator, ensures communication cost is recorded identically for both, and verifies that the implementation matches the formal model.
- **Thaker Shivangi Pareshkumar** — experimental design and analysis: organises the experiments over communication budget, sensor reliability, team size, and building complexity, keeping comparisons consistent; analyses the four metrics, presents the results, documents observed trade-offs and limitations, and relates the findings back to the research question.

All members will contribute to system integration, testing, debugging, report writing, and preparation of the final submission and presentation.

6 Conclusion

This project will investigate how a limited communication budget affects coordination and task allocation among heterogeneous agents in a simulated search-and-rescue setting, where the agents that can find the victim are not the agents that can rescue them. Task allocation is modelled using constraint satisfaction and distributed constraint optimisation, and a centralised baseline is compared with a decentralised DCOP-based method across communication budgets, sensor reliability levels, team sizes, and building complexities. The intended outcome is a quantitative account of the trade-off between communication cost and rescue performance, together with the conditions under which decentralised coordination remains effective and a clear statement of the limitations of the simulation.

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